

Appendix to “SCOUT: Semantic Coverage-Guided Scenario Generation for Autonomous Driving System Testing”

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This appendix is a separate document accompanying the anonymous artifact repository. Sections, tables, and figures use A-series numbering so that the main paper can reference them unambiguously.

1 TARGET DSL Extensions

Table 1 formally defines the scenario DSL implemented in the artifact. The DSL follows the modular DSL-to-scenario generation pattern exemplified by TARGET [1]: a scenario program is compiled to an internal ScenarioSpec and, where custom geometry is required, to an OpenDRIVE map. The constructs below are the extensions added for semantic coverage-guided testing (actor controllers, semantic targets, and the parameterized five-dimensional search space).

2 Hazard Templates

The artifact defines nine hazard templates (Table 2). Templates are developer-defined conjunctions over the data-derived predicate vocabulary; each template is instantiated automatically by matching it against the nuScenes STSG corpus (Section A.3), yielding the hazard obligations of the semantic obligation inventory. Formulas follow the DoTA-style temporal notation used in the inventory artifact where available.

3 Implementation Details

Hazard instantiation. For each scene slice, the matcher first identifies graph nodes satisfying a template’s node predicates, then incrementally verifies the associated attribute, relation, and temporal constraints. Every complete match corresponds to one instantiated hazard obligation, recorded in the inventory with its supporting slices. Constituent predicates are entirely data-derived; the templates themselves are versioned and auditable.

Search algorithm. The search is an elitist hill-climber. A single candidate parameter vector is evaluated per step; after the execution, the coverage engine updates the suite-level obligation set. For the semantic policy, a candidate’s fitness is the number of previously uncovered obligations its execution closes (ties broken by the total number of witnessed obligations; collisions are never rewarded). The next candidate is a Gaussian mutation of the current elite within the five-dimensional

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Table 1. Scenario DSL constructs implemented in the artifact.

Construct	Fields	Semantics
[environment]	town, weather, tick_hz	Simulation map, weather preset, and stepping rate (10 Hz in this study).
[ego]	route	Ego route anchor (spawn and goal spawn-point indices); the ADS under test drives the ego.
[[actor]]	type, spawn, controller, trigger, speed_ms	Adversarial actor with an explicit controller binding and a distance- or tick-based trigger.
[targets]	obligations, hazard_class	Semantic targets for the coverage engine. Targets are metadata for generation; coverage credit always flows from observation.
controller_param	trigger_radius_m, speed, spawn_transform, destination_location, trigger_tick	Parameters mutated by the search: trigger radius, actor speed, lateral and longitudinal staging offsets, and release tick.

Table 2. All nine hazard templates with formal definitions.

Template	Semantic definition	Temporal formula (inventory artifact)
other: ahead_or_waiting	vehicle actor that is in front of the ego and stationary for ≥ 2 consecutive keyframes	Eventually(in_front_of(adversary, ego) AND (stationary(adversary) OR braking(adversary)))
other: start_stop_or_stationary	vehicle actor that is in front of the ego and drops ≥ 1.0 m/s of speed across the slice	Eventually(in_front_of(adversary, ego) AND stationary(adversary))
other: oncoming	vehicle actor with opposed heading in front of the ego that is closing distance	Eventually(oncoming(adversary) AND approaching(adversary, ego))
other: lateral	vehicle actor that changes to a laterally adjacent lane and crosses the ego path	Eventually(lane_changing(adversary) AND crossing_path(adversary, ego))
other: turning	vehicle actor that turns ≥ 30 deg while crossing the ego path	Eventually(turning(adversary) AND crossing_path(adversary, ego))
other: obstacle	static movable object on the road, in front of the ego, in the ego lane	Eventually(on_road(adversary) AND obstructing(adversary, ego))
other: pedestrian	pedestrian on the roadway outside a ped_crossing polygon that crosses the ego path (jaywalking proxy)	Eventually(jaywalking(adversary) AND crossing_path(adversary, ego))
pedestrian_in_path	EXP-018 explicit class: pedestrian in front of the ego whose swept path crosses the ego corridor	–
oncoming_cut_in	EXP-018 explicit class (paper example): oncoming actor that reaches the ego lane later in the slice	–

space; the random policy samples uniformly, and the LSA and KMNC policies mutate around the elite under their respective fitness values. Near-duplicate candidates are rejected before execution. The five dimensions and bounds are: trigger radius [5, 35] m, actor speed [0.8, 4.0] m/s, lateral staging offset [1, 14] m, longitudinal staging offset [−10, 30] m, and trigger tick [10, 400] at 10 Hz. Each hazard template carries its own parameter space (pedestrian crossing: trigger radius, walker speed, lateral and longitudinal staging offsets, trigger tick; lead-vehicle braking: lead gap, trigger radius, brake strength, pre-brake throttle), and an obligation scheduler selects the highest-priority uncovered obligation, routes it to its template, and advances to the next uncovered obligation once the current one is realised.

Table 3. Semantic obligation inventory (complete nuScenes trainval split).

Property	Value
Scenes processed	850
Scene slices	15,204
Defined predicates	38
Node / attribute / relation / hazard obligations	8 / 51 / 35 / 9
Total instantiations	103
Slices per obligation (median)	1,892
Obligations witnessed by <5 slices	3

Table 4. Per-map endpoint relation and hazard coverage, InterFuser campaign (mean over routes; cells show Cov_E / Cov_H).

Map (routes)	Semantic	Random	LSA	KMNC
Town01 (6)	0.182 / 0.222	0.172 / 0.204	0.206 / 0.244	0.172 / 0.204
Town03 (2)	0.188 / 0.222	0.188 / 0.222	0.188 / 0.222	0.188 / 0.222
Town05 (3)	0.167 / 0.222	0.188 / 0.222	0.167 / 0.222	0.167 / 0.222
Town10HD (3)	0.115 / 0.222	0.115 / 0.222	0.115 / 0.222	0.125 / 0.222

Table 5. Per-map endpoint relation and hazard coverage, AutoVLA campaign (mean over routes; cells show Cov_E / Cov_H).

Map (routes)	Semantic	Random	LSA	KMNC
Town01 (4)	0.188 / 0.222	0.177 / 0.222	0.188 / 0.222	0.188 / 0.222
Town03 (2)	0.188 / 0.222	0.156 / 0.222	0.188 / 0.222	0.188 / 0.222
Town05 (3)	0.188 / 0.222	0.188 / 0.222	0.188 / 0.222	0.172 / 0.167
Town10HD (3)	0.125 / 0.222	0.172 / 0.222	0.167 / 0.222	0.146 / 0.222

Coverage engine. The engine matches observer-emitted facts against the inventory through a declarative crosswalk. Of the 103 trainval obligations, 78 (75.7%) are mapped, each labelled direct, derived, or proxy; unmapped obligations are recorded with reasons, and four predicate types (colliding, occluded, out_of_control, speeding) are explicitly ungrounded. Coverage is reported per dimension with first-witness ticks and provenance; hazard conjunctions are credited only within a single trace.

Statistics. Policy comparisons use the Wilcoxon signed-rank test over per-route paired differences with Holm–Bonferroni correction across the three comparisons, Vargha–Delaney $\hat{A}_{12}$ effect sizes, and bootstrap 95% confidence intervals (2,000 resamples, seed 0). Seeds are paired across policies per route (seed 13).

TARGET pipeline figure. Figure 3 of the TARGET paper [1] illustrates the modular DSL-to-scenario pipeline that this work follows. For copyright reasons the figure is not reproduced here; readers are referred to the original publication.

Table 6. Failure-causing semantic obligations and instances over failing executions (both ADSs pooled). An instance is a distinct (obligation, trigger-radius 5 m bucket) pair.

Policy	Failing executions	Obligations	Instances
Semantic	529	21	131
Random	511	21	147
LSA	546	29	155
KMNC	506	21	147

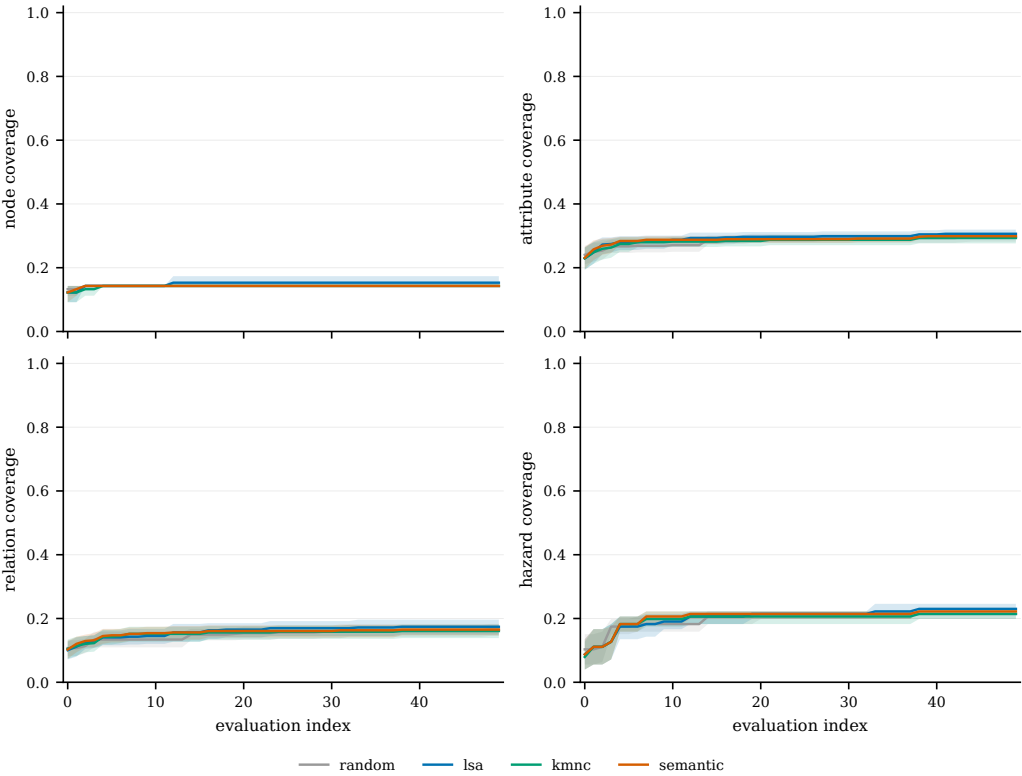


Fig. 1. Coverage growth by dimension, InterFuser campaign (Figure A1).

4 Additional Tables and Figures

References

[1] Yao Deng, Zhi Tu, Jiaohong Yao, Mengshi Zhang, Tianyi Zhang, and Xi Zheng. 2025. TARGET: traffic rule-based test generation for autonomous driving via validated LLM-guided knowledge extraction. *IEEE Transactions on Software Engineering* 51, 7 (2025), 1950–1968.

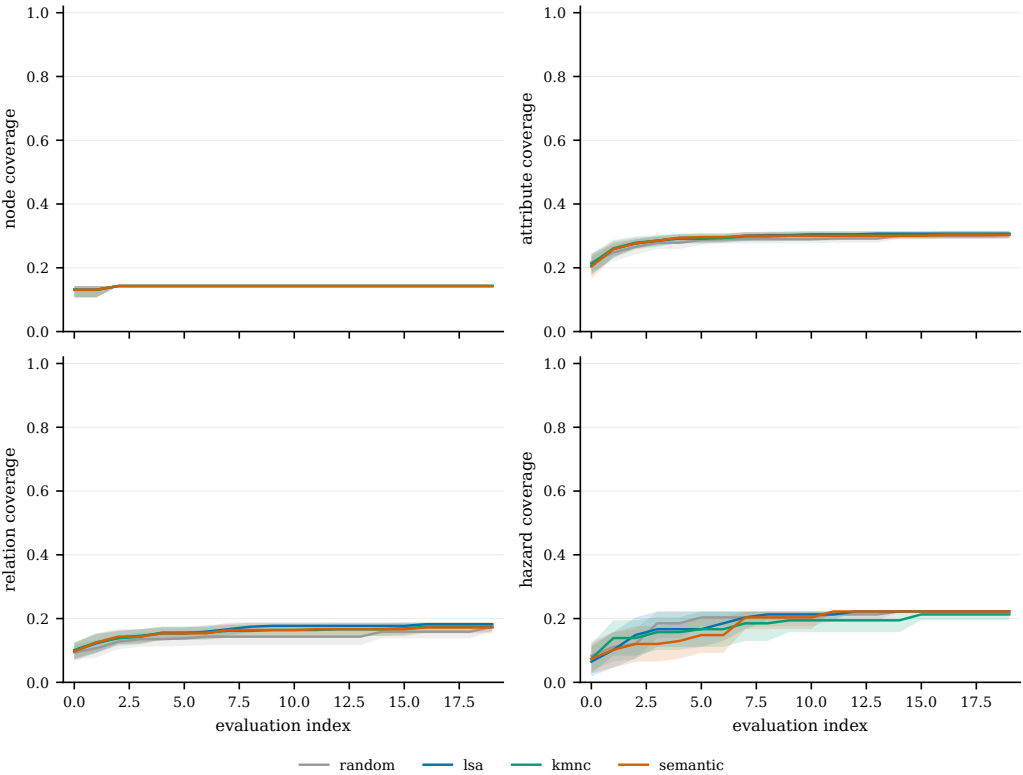


Fig. 2. Coverage growth by dimension, AutoVLA campaign (Figure A2).